CSC 220.03

Arpita Misal, SFSU ID: 921644032

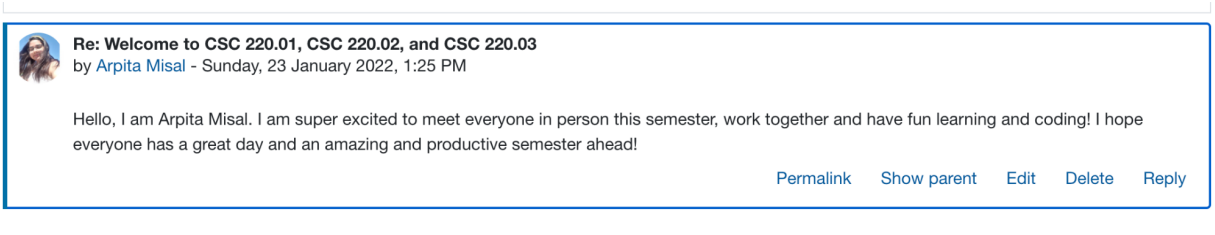
Assignment 01

Due Date & Time: 01-31-2022 at 11:55 PM

Assignment 01 Report

PART A - Communication

1. iLearn



2. Emailing
   - YES. This is to confirm that I will use my SFSU email address when contacting my grader and my course instructor so that my emails will not be filtered. Thank you.
   - YES. This to confirm that when I contact my course instructor, I will start my email subject with this format "CSC 220.03 SHAM" so that I will get an answer timely. Thank you.
   - YES. This is to confirm that if I do not get an answer within 24 hours, I will check if I sent my email properly. And in either case, I will kindly resend my email message. Thank you.
3. Grader

   Question 1: What is the URL of your grader's forum on iLearn?

   - .03 (SHAM) - TuTh 09:30 AM - 10:45 AM

Question 2: What is your grader's full name?

- PeiJun Huang

Question 3: What is your grader's SFSU email address? Please email your grader when you have questions.

- phuang5@mail.sfsu.edu

4. Guidelines for All assignment, Assignment Report Template, and all future guidelines
    - YES. This is to confirm that I have carefully read, understood, and agreed to the Guidelines for ALL Assignments above and the Assignment Report Template. I will also carefully read, understand, and agree to any additional guidelines which the instructor and the grader(s) will provide. I will strictly follow all the instructions.
5. Course Policy on Student Conduct and Academic Honesty
    - YES. This is to confirm that I have carefully read, understood, and agreed to the Course Policy on Student Conduct and Academic Honesty which was distributed to me with the course syllabus and whose digital copy was shared with me on the File Manager. I am acutely aware that the policy includes, but is not limited to, the San Francisco State University's Code of Student Conduct (at https://conduct.sfsu.edu/standards), the Computer Science Department's Student Policies (at https://cs.sfsu.edu/student-policies), and the Honor Code of this course (at http://csc220.ducta.net/00-README-StudentConduct_AcademicHonesty.pdf). I will strictly follow all the rules.
6. CSC220 File Manager

Advising for CSC220 Students (Spring 2018, updated Spring 2021):

By Dr. William Hsu, *Department Chair, Computer Science Department*
and Jennifer Schwartz, *Office Manager, Computer Science Department*

1. CSC220 students should check our prerequisite flowchart for progress through the major: http://cs.sfsu.edu/undergrad/under-prereq.html

2. It's a good idea to take 220 and 230, and also Math 227, concurrently. Then they can progress on to CSC 340 next semester.

3. The longest dependent chain of CS courses is 210-220-340-413-600. A student should try to complete one course in the chain every semester.

4. A full-time course load is 12-18 units. This means 40 hours of course work per week, including programming assignments, studying, reading etc. If a student has a part-time job, s/he should scale back the course load accordingly. For example, a student who is working 20 hours per week should ideally take no more than 9 units that semester.

5. In each semester, a student should take a mix of CS major courses and GE courses. CS major courses will take more time and effort; it's important not to fill up one's schedule with just CS major courses. For example, a typical schedule for a student in 220 would be CSC 220 (3 units), CSC 230 (3 units), Math 227 (4 units), and a GE course or two (3-6 units).

6. Based on (5), students should not take all their GEs first! They should save some GEs, to space out their CS major courses every semester.

- Please also watch our welcome video: https://www.youtube.com/watch?v=QWXu_jqkxt8

CSC220/StrongFoundationCSC210:

IDX	FILE NAME	TYPE	SIZE	LAST MODIFIED	PERMS
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3	2019Fall-CSdept-AddendumToCourseSyllabi.pdf	File	0.05 MB	08:29 01-03-22	604
4	2019Fall-CSdept-CSC210Desc.pdf	File	0.10 MB	08:29 01-03-22	604
5	2019Fall-Calendars_Academic.html	File	0.00 MB	08:29 01-03-22	604
6	2019Fall-Calendars_Final_Examination.html	File	0.00 MB	08:29 01-03-22	604
7	2019Fall-RecommendedServices.html	File	0.00 MB	08:29 01-03-22	604
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CSC220 iLearn: Announcements and Forums

PART B - NetBeans IDE Installation

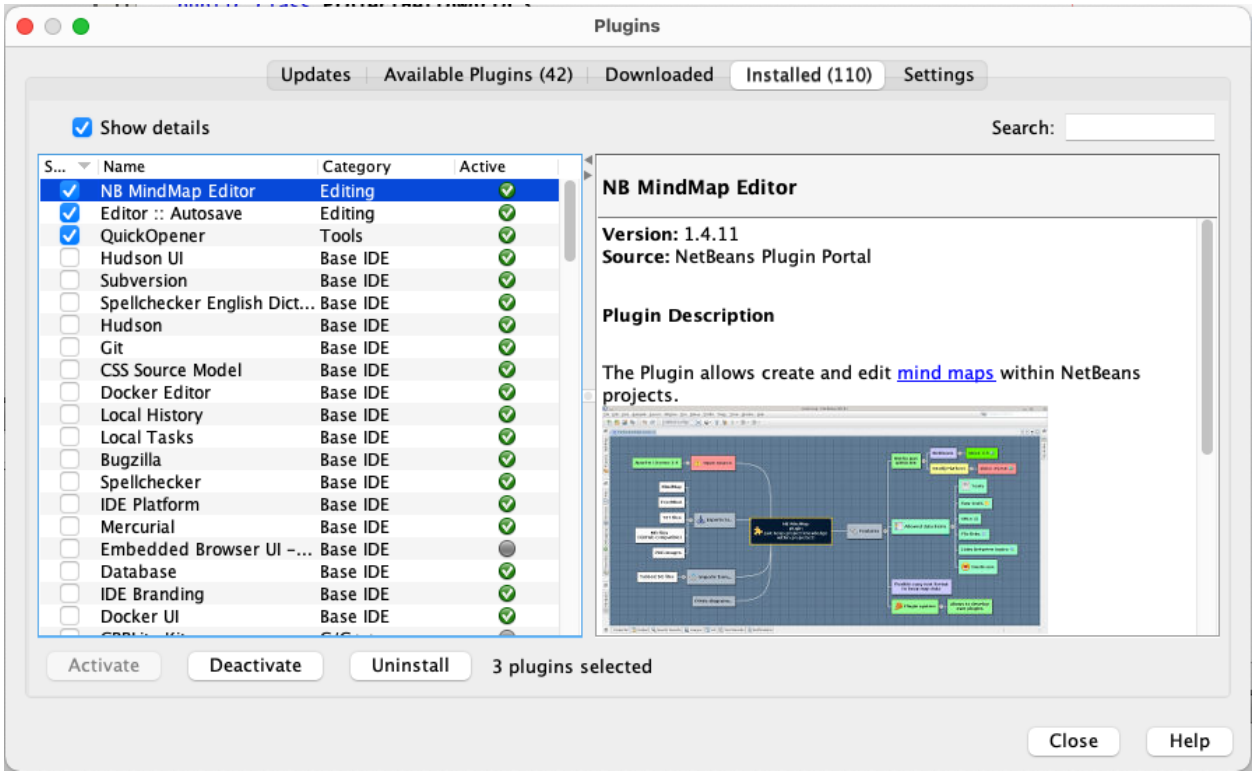
```java
 * CSC 220.03
 * Arpita Misal, SFSU ID: 921644032
 * Assignment 01
 * Due Date & Time: 01-31-2022 at 11:55 PM

 */

import java.util.*;
public class MyFavoriteApp {
   public static void main(String[] args)
    {
        int[] A=null;
        int size;

        Scanner input = new Scanner(System.in);
        System.out.println("Enter size of array: ");
        size = input.nextInt();
        A = new int[size];
        System.out.println("Enter " + size + " integer data: ");
        for(int i = 0; i < size; i++){
            A[i] = input.nextInt();
        }

        //call display() method to display array data
        System.out.println("======================");
        System.out.println("Array values:");
        display(A);
        System.out.println("======================");

        // Selection sort algorithm to sort data in A
        selectionSort(A);

        //display sorted array data
        System.out.println("Sorted array values:");
        display(A);
    }
    // display content of integer array
    static void display(int[] B) {
        for (int i=0; i<B.length; i++) {
            System.out.println("array value:"+i+":"+B[i]);
        }
    }

    // Sort values in B using Selection sort method
    static void selectionSort(int[] B) {
        int n = B.length;
        for (int i=0; i<n-1; i++) {

            int x=findMinimumIndex(B, i, n);

            swapData(B, i, x);
        }
    }

    // find index with minimum value in array B[i...n)
    static int findMinimumIndex(int[] B, int i, int n) {
        int min = i;

        n = B.length;
        for(int j = i + 1; j < n; j++){
            if(B[j] < B[min]){
                min = j;
            }
        }
        return min;
    }

    // Swap B[i] and B[j]
    static void swapData(int[] B, int i, int j) {
        int temp = B[i];
        B[i] = B[j];
        B[j] = temp;
    }
}
```

```
run:
Enter size of array:
4
Enter 4 integer data:
12
3
5
2
======================
Array values:
array value:0:12
array value:1:3
array value:2:5
array value:3:2
======================
Sorted array values:
array value:0:2
array value:1:3
array value:2:5
array value:3:12
BUILD SUCCESSFUL (total time: 15 seconds)
```

There are a few problems in this program that can be made better. One is that there are not enough comments or documentation to give another person a clear idea of what is happening with that specific line or block of code. So, I can make it better by adding more comments in this program. Another thing is that the array name and a few other data members are not named properly for another person to understand. I am the only person who knows what A or n might mean but not others. Thus, I need to name the variables with its particular meaning. Also, if I want to go above and beyond to make my program even more convenient and better for the user, I can use an ArrayList rather than an array which is fixed in size and capacity. The user always has to enter the number of terms that are going to be in the list but if we use an ArrayList we will be able to add, remove, set and get the number of terms based on the user's wish. It is flexible and much easier to work with than an array which might create problems in this kind of situation. up th eIt is also much easier to the user if they chose a term in the beginning but realized they wanted more or less terms. Thus rather than running the whole program and typing all the terms, using an ArrayList will make it more efficient. These are the three things that I would add to my program to make it better and more efficient.

PART C - NetBeans Plugins and Shortcuts

1. Install 3 Netbeans plugins



- NB MindMap Editor: This plugin lets me edit and create mind maps within any NetBeans projects. I decided to install this plugin because it would help me understand my programs better. I am a person who does better once I get a visual of something I want to do, be it with either mind maps or diagrams. Therefore, this plugin will help me do that.
- Editor :: Autosave: I decided to install this plugin so that if I forget to save it can autosave. There are many times when I don't have my full attention on the program that I am writing, this plugin can be especially helpful in these moments. In this class when we write and do projects, we will write multiple lines of code.

There is a lot of effort put into that, therefore in order to not lose that data it is important to save my work.

2. Netbeans-shortcuts-80.pdf
    a. My favorite shortcuts are the code templates, especially pvsm and sout. These two are the shortcuts that I use the most. In the beginning stages of my coding experience I had to repeatedly write the print statements and the main method signature. At that time, these shortcuts saved a lot of time and effort.

PART D - Class Design Guidelines

It is extremely important to convey to someone what you are doing while writing your code. If someone who knows nothing about what you are writing your program on comes and checks your work they should be able to follow your thinking and code. You can accomplish this by consistency, a class design guideline which focuses on consistency of naming conventions, constructors and other methods throughout all classes and programs. It is good practice to follow the proper naming conventions to identify what type of value it holds. For example, a variable has the first word lowercase and the next word's first letter uppercase(helloWorld) and a class name has all the words' first letter uppercase(HelloWorld). It is also important to write comments to explain your code. We have to document everything to communicate with others. This is extremely important when multiple people are working on one project. In this case it would be a problem if one person does not understand what you have written, it would be a mess essentially. When I started coding, it was annoying to comment and document my code but I realized it's importance when I did a program long ago and when I came back to it after a long time, I was unable to remember what this was about. Therefore, it is

extremely important to document and be consistent throughout your classes and projects.

Let me give you a piece of code to show its demonstration:

```
// Imports the Scanner class from the java util package.(COMMENTS)
import java.util.Scanner;
// Creates public class KnockKnock(COMMENTS/Description)
public class KnockKnock { (Pascal case naming convention)
   // main method(COMMENTS/Documentation)
   public static void main(String[] args){
       // Declares and initializes input as the predefined(COMMENTS)
       // input from the user.(COMMENTS)
       Scanner input = new Scanner(System.in);(Camel case naming convention)
       System.out.println("Knock, knock");
       // Takes input from the user.(COMMENTS)
       String answer1 = input.nextLine();
       System.out.println("Annie");
       // Takes input from the user.(COMMENTS)
       String answer2 = input.nextLine();
       System.out.println("Annie thing you can do I can do better");
   }
}
```